

Week 12: Chapter 10 – Establishing Requirements

A point made in this chapter that I found quite interesting was that the same keyword for a desirable characteristic (higher-level functional requirement) in highly different contexts can mean wildly different things even though they are characterized by the same keyword. The book uses the example that describes how “trustworthiness” in an online shopping site vs. a robot companion is parameterized in fundamentally different ways. Trustworthiness in the shopping site means robust cybersecurity, whilst trustworthiness in a robot companion is about the robot being able to behave like a human whose behavior makes other humans perceive them as trustworthy. Because one case regards human perception of trustworthiness, while the other is a very technical feature that involves no human perception, I found the example to be really great in showing how a factor can be considered in radically different ways.

Section 10.3 talks about environmental (user) requirements. It lists the types of circumstances to consider as physical, social, organizational, and technical. I feel that categorizing the types of environmental requirements in this manner is optimized for workplace and professional task-oriented interaction context. What if the interaction context is more personal, like family or mental health-oriented? Interaction design for these contexts very clearly exist. The type of social environment accounted for in the book example for social environment requirements is not very complex, such as asynchronous vs. synchronous data sharing. The type of environment considered is very purely practical, even when looking at social factors. But when we are looking at family interaction, or product design for individual mental health, we have to account for more complex social factors or subcategories such as emotions, the nature and quality of interpersonal relationships, social distance between individuals, as well as empathy and personality traits. We also must consider individual mental state for mental health applications as a form of “user’s internal environment”, or the way that the user *perceives* a given environment.

When I was reading about the data gathering guidelines for requirements, one of the guidelines confused me and I wasn’t sure how it can be implemented well. It was the guideline that emphasized the necessity of working with all stakeholder groups and in particular involving “all of the right people”. But how can we ensure that *all* of the right people are identified? It seems that in most industrial product development processes it’s next to impossible to involve everyone that is important. In some way everyone is important, but on a large-enough scale there are so many different stakeholders who can potentially be involved that it’s difficult to ensure all of the important people are involved. Say, in the example presented by the textbook, they said that a system caused operation to fall by 50% because people didn’t find that the supplier warehouses had a major part in the process. If they were really key, they should have been easily accounted for in the first place. But if you are only responsible for one small department in a company, but depend on people from other departments for this type of information, and they overlook something that is actually key to your product development context, it can cause huge

problems that are difficult for your department alone to prevent or deal with. I just wonder if there is a clear way to mitigate this kind of problem.